Maritime AI Threat Detection for Moving Ships

Compiled discussion: ML integration, datasets, ship/iceberg SAR classification, cyclone detection and forecasting, real-time threat assessment, CPA/TCPA, routing, MarineChain, and implementation steps

Purpose. This document consolidates the technical discussion from this chat into a single implementation-oriented reference. It preserves the main ideas, equations, workflows, cautions, and research directions discussed.

1. Initial Cyber-Resilient PV Paper Direction

The first proposed direction was an AI-enabled cyber-resilient solar PV forecasting framework. The central idea was to inject cyberattacks into PV/weather measurements, detect corrupted inputs, reconstruct them, and preserve forecasting performance.

```
PV/weather data -> cyberattack injection -> attack detection -> data reconstruction ->
                        forecasting -> resilience assessment
```

Suggested attack models included false-data injection, persistent bias, multiplicative scaling, replay attacks, and communication/data-loss attacks. A denoising autoencoder or Isolation Forest could provide anomaly detection, while a Transformer could provide PV forecasting.

The key methodological lesson carried forward into the maritime work is that AI should not merely classify data: it should be embedded in a complete cyber-physical pipeline that detects abnormal conditions, estimates risk, and supports an operational response.

2. ML Connection to MarineChain / PoAh

The maritime extension proposed keeping the existing blockchain, Proof-of-Authentication (PoAh), ABAC, ECIES/ECDSA, geofencing, and trusted communication mechanisms, while adding ML as an intelligent anomaly/threat-detection layer.

```
Sensor -> Vessel -> HAPS -> ML Threat Detector -> Control Station -> PoAh -> Blockchain
```

ML can evaluate behavioral and sensor-data patterns, while cryptographic mechanisms evaluate identity, integrity, authorization, and provenance. This separation avoids treating ML as a replacement for security.

2.1 Suggested ML tasks

• Binary attack detection: Normal vs Attack.

• Multiclass detection where labels permit: GPS spoofing, false-data injection, replay, DoS, impersonation, etc.

• Malicious-validator risk estimation.

• AIS/GPS trajectory anomaly detection.

• Sensor-data anomaly detection and reconstruction.

2.2 Suggested models

• Logistic Regression - simple baseline.

• Decision Tree - interpretable baseline.

• Random Forest - conventional ensemble baseline.

• XGBoost - strong tabular classifier.

• Autoencoder - anomaly/unknown-attack detection.

• LSTM - optional sequence-based detector.

2.3 ML-assisted trust score

```
TS_i = VB_i + ST_i - MV_i

p_i = P(malicious | x_i)

TS_i^ML = alpha * TS_i + (1-alpha) * (1-p_i)
```

The ML probability can therefore supplement, rather than replace, the deterministic PoAh trust logic.

3. Dataset Strategy

3.1 Cybersecurity data

For network-level cyberattack detection, a labelled IoT intrusion dataset can train RF/XGBoost/LSTM models. CICIoT2023 was suggested as a benchmark family, with the important limitation that it is IoT benchmark traffic rather than native maritime traffic.

3.2 AIS data

AIS data is appropriate for vessel behavior and GPS/trajectory anomaly detection. Useful fields include timestamp, vessel identifier/MMSI, latitude, longitude, speed over ground (SOG), course over ground (COG), and heading.

3.3 Statoil/C-CORE SAR data

The uploaded Statoil/C-CORE guide describes ship-vs-iceberg classification using two SAR polarization bands. Each band contains 5,625 values that are reshaped into a 75 x 75 image. The training target is is_iceberg: 1 for iceberg and 0 for ship.

```
band_1 -> 75 x 75 | band_2 -> 75 x 75

band_3 = (band_1 + band_2) / 2

X = [HH, HV, average] -> 75 x 75 x 3
```

The guide compares Custom CNN, ResNet50, EfficientNetB0, MobileNetV2, and DenseNet121 under a common validation protocol.

3.4 Cyclone data

Cyclone detection and trajectory prediction require a separate meteorological dataset. Useful fields include timestamp, cyclone latitude/longitude, maximum wind, central pressure, movement speed/direction, sea-surface temperature where available, and historical track information.

4. Extension: Cyclone-Aware Ship Safety

The cyclone extension should not be implemented by adding a third class to the ship-vs-iceberg classifier. The SAR dataset has no cyclone labels, wind fields, pressure, or cyclone tracks. A multimodal system is therefore required.

```
SAR branch:
SAR HH/HV -> CNN/EfficientNet -> Iceberg probability

Weather branch:
Cyclone history + weather -> LSTM/BiLSTM/Transformer -> Cyclone track

Navigation branch:
AIS/GNSS -> Ship state and future trajectory

Fusion:
Iceberg + Cyclone + Ship state -> Threat engine -> Alert / Safe route
```

5. Real-Time Architecture

```
SHIP / VESSEL
|
+-- GNSS/AIS: latitude, longitude, SOG, COG, heading
```

```
+-- Radar/SAR/Camera
+-- Weather / onboard sensors
|
v
EDGE COMPUTER
|
+-- Iceberg AI
+-- Cyclone monitor / forecast
+-- Ship trajectory estimator
|
v
THREAT ENGINE
|
+-- Distance
+-- CPA
+-- TCPA
+-- Severity / probability
|
+--> Local warning
+--> Dynamic route planner
|
v
HAPS / Satellite -> MarineChain / PoAh -> Control Center
```

A key design decision is to keep the fast onboard safety loop independent of blockchain consensus. Immediate hazard detection and route warnings should run locally; MarineChain should authenticate, record, and transmit the resulting warning and state information.

6. Moving Ship State

Every GNSS/AIS update forms the current vessel state:

```
S_t = [Lat_t, Lon_t, SOG_t, COG_t, Heading_t, time_t]
```

A research prototype can replay historical AIS observations at a fixed interval so that the ship icon moves continuously on a digital map. This creates a realistic streaming environment without requiring an actual vessel.

7. Iceberg Detection and Tracking

The Statoil model can first be trained offline as a SAR image classifier:

```
P_I = P(iceberg | X_SAR)
```

However, classification alone does not provide bearing or distance. A real deployment needs sensor-specific object localization and tracking. For the initial simulation, classified iceberg samples can be associated with clearly identified simulated positions.

Once a geographic or relative track is available, represent the iceberg state as:

```
I_t = [Lat_i, Lon_i, V_i, theta_i]
```

Successive observations allow estimation of whether the iceberg is drifting toward or away from the vessel.

8. Cyclone Detection and Forecasting

Cyclones should be monitored from meteorological/satellite information before they become locally observable hazards.

```
C_t = [Lat_c, Lon_c, Wind, CentralPressure, MoveSpeed, Direction]
```

A sequence model can predict future cyclone positions:

```
C_hat_(t+1:t+H) = f(C_(t-k:t))
```

Suggested models are LSTM, BiLSTM, and Transformer. The predicted track should be updated whenever new observations arrive.

9. Distance, CPA and TCPA

Distance alone is insufficient because both the ship and hazard can move. The system should estimate the closest future encounter.

```
D_h = distance(S_hat_(t+h), C_hat_(t+h))

D_CPA = min_h D_h

T_CPA = argmin_h D_h
```

The same concept can be used for tracked icebergs. CPA means Closest Point of Approach; TCPA is the time to that closest approach.

9.1 Haversine distance

```
d = 2R * asin( sqrt( sin^2(dphi/2) + cos(phi_s)cos(phi_h)sin^2(dlambda/2) ) )
```

10. Threat Engine

Maintain hazard-specific risk indicators rather than immediately forcing every hazard into one arbitrary score.

```
R_I = f(P_iceberg, Distance_I, CPA_I, TCPA_I)

R_C = f(Wind, Pressure, Distance_C, CPA_C, TCPA_C)
```

Experimental risk levels can be Low, Moderate, and High, but any thresholds used in a research prototype should be explicitly described as experimental unless they are derived from maritime standards or validated operational guidance.

10.1 Optional combined research score

```
R_total = alpha * R_iceberg + beta * R_cyclone + gamma * R_cyber
```

Weights should be calibrated, optimized, or evaluated through sensitivity analysis rather than selected solely to improve reported results.

11. Dynamic Route Planning

Once hazards are localized and forecast, the next operational step is route recommendation.

```
G = (V, E)

Cost(e) = alpha*Distance(e) + beta*IcebergRisk(e) + gamma*CycloneRisk(e) + delta*WeatherRisk(e)

Route* = argmin_R sum_{e in R} Cost(e)
```

A* is a practical first algorithm. The route should be recalculated whenever hazard forecasts materially change.

12. Real-Time Dashboard

The dashboard should display the moving vessel, planned route, iceberg locations/tracks, cyclone center and predicted path, current distance, CPA/TCPA, wind/intensity information, risk level, and active route warning.

```
MARITIME THREAT MONITOR
Ship: moving icon + heading + speed
Iceberg: probability + distance + CPA/TCPA
Cyclone: current center + forecast track + intensity + CPA/TCPA
Risk: Low / Moderate / High
Route: current path + dynamically recomputed safe path
Alerts: time-stamped warning panel
```

13. Streaming Software Architecture

A manageable prototype stack is Python with TensorFlow/PyTorch for AI, MQTT or WebSocket for streaming updates, FastAPI for inference services, a time-series/database layer for telemetry, and a web map/dashboard for visualization.

```
AIS Simulator --------Weather/Cyclone Feed ---+--> MQTT/WebSocket --> AI Services
SAR/Radar Simulator ----/ |
+--> Iceberg model
+--> Cyclone model
+--> Ship predictor
|
v
Risk Engine
/ Dashboard MarineChain
```

14. MarineChain Integration

The recommended split is:

```
Fast safety loop: Sensors -> AI -> Threat Engine -> Alert / Route

Trusted communication loop: Threat result -> HAPS -> PoAh -> Blockchain -> Control Center
```

A hazard message can contain Threat_ID, Ship_ID, timestamp, position, threat type, probability, CPA, TCPA, risk level, model identifier, and digital signature.

15. Cyberattack Extension

After the normal hazard system works, cyber-resilience experiments can manipulate ship GPS, cyclone coordinates, wind measurements, or hazard messages.

```
Lat' = Lat + delta_lat

Lon' = Lon + delta_lon

 Wind' = alpha * Wind
```

The research question then becomes whether the system can continue to provide trustworthy maritime warnings when physical hazards and cyber manipulation occur together.

16. Recommended Step-by-Step Build Order

1. Define outputs: moving vessel, iceberg risk, cyclone risk, CPA/TCPA, warning, and safe route.

2. Create the project folders and common data schema.

3. Obtain an AIS trajectory and replay it as a moving ship.

4. Build the live map/dashboard and verify ship motion.

5. Load and preprocess the Statoil/C-CORE SAR data.

6. Train the five ship-vs-iceberg models under the same split and preprocessing.

7. Evaluate Accuracy, Precision, Recall, F1, ROC-AUC, and Log Loss; save the selected model.

8. Replay a historical cyclone track on the same map.

9. Calculate real-time ship-to-cyclone distance.

10. Predict the ship's future motion using a simple constant-course/constant-speed baseline.

11. Train LSTM/BiLSTM/Transformer for cyclone trajectory prediction.

12. Compute cyclone CPA and TCPA.

13. Add iceberg position/tracking simulation and compute iceberg CPA/TCPA.

14. Build the hazard-specific threat engine.

15. Implement Low/Moderate/High experimental warnings.

16. Add A* dynamic route planning around evolving hazard zones.

17. Connect components using MQTT/WebSocket streaming.

18. Integrate MarineChain/HAPS/PoAh for authenticated hazard-message transmission.

19. Inject GPS/weather/message attacks and evaluate cyber resilience.

20. Run ablation studies: AI only vs AI+risk engine vs AI+routing vs AI+routing+MarineChain.

17. Minimum Viable Prototype

The first working prototype should be deliberately small:

```
Historical AIS replay + historical cyclone replay + live map + distance + CPA/TCPA
```

Only after this works should the iceberg AI, cyclone prediction model, dynamic routing, MarineChain, and cyberattack experiments be added. This minimizes integration risk and creates demonstrable milestones.

18. Suggested Research Progression

```
Paper 1: Ship-Iceberg SAR Classification

Paper 2: Cyclone-Aware Moving-Ship Risk Prediction

Paper 3: Multi-Hazard + Cyber-Resilient MarineChain
```

For the cyclone paper, the stronger contribution is not merely cyclone detection. It is cyclone trajectory prediction relative to a moving vessel, closest-approach assessment, risk estimation, and dynamic route adaptation.

19. Important Experimental Cautions

• The Statoil/C-CORE dataset is a SAR ship-vs-iceberg classification dataset; it is not a live onboard radar stream.

• Cyclone prediction requires a separate meteorological/track dataset.

• AIS and cyclone datasets need not represent the same historical event for an initial simulation, but synthetic encounters must be labelled as simulations.

• Do not claim real-world safety performance from simulated threat thresholds.

• Do not select a model only by accuracy; inspect precision, recall, F1, ROC-AUC, calibration/log loss, and class imbalance.

• Keep the immediate onboard warning loop independent of blockchain latency.

• Report inference latency and update rate if claiming real-time capability.

• For stronger validation, later synchronize real AIS tracks with real cyclone tracks and sensor observations.

20. Final Integrated Concept

```
Moving Ship + Real-Time Sensing + Iceberg AI + Cyclone Prediction + CPA/TCPA + Dynamic Routing +
                                        MarineChain
```

This architecture transforms a static image-classification exercise into a maritime cyber-physical safety platform. The ship continuously moves, hazards are independently observed and predicted, encounter geometry is evaluated, warnings and routes are updated, and MarineChain provides a trusted communication/audit layer.